Experiment 20

STFT and Wavelet-Based Time-Frequency Analysis of Non-Stationary Signals

OBJECTIVES

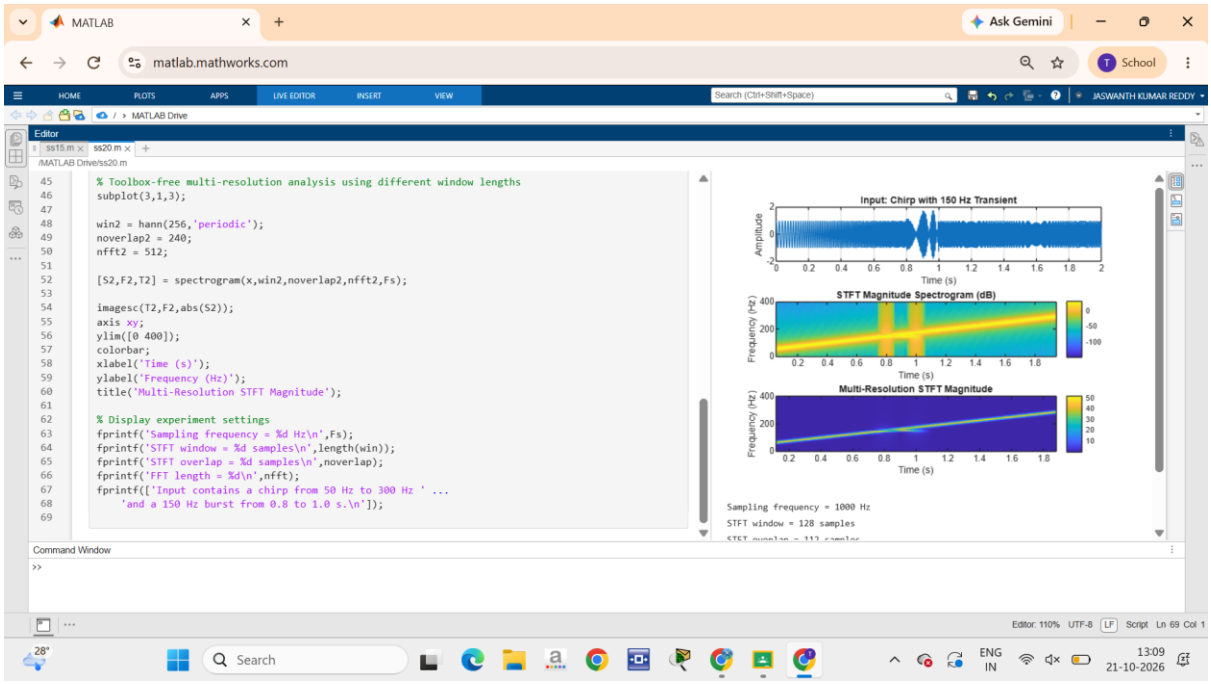
• To generate a non-stationary signal containing time-varying frequency components and a transient.

• To compute and visualize the STFT spectrogram.

• To perform wavelet-based time-frequency analysis and interpret the scalogram.

• To compare STFT and wavelet representations for identifying transient and changing-frequency components.

Output Graphs:



Result: A non-stationary signal containing a frequency-varying chirp and a transient component was analyzed using the short-time Fourier transform (STFT) and continuous wavelet transform (CWT). The spectrogram and scalogram were used to observe time-varying frequency components.